

Dhrumil Bhut

Software Engineer | Backend Systems & API Development

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PROFESSIONAL SUMMARY

Software Engineer with 2.5+ years of experience building scalable backend systems using Node.js, PostgreSQL, and AWS. Proven track record of designing APIs, optimizing databases, and delivering production-grade infrastructure. Additionally skilled in Applied AI and LLM systems, including RAG pipelines, multi-agent workflows, and MLOps.

WORK EXPERIENCE

Software Engineer

08/2023 – Present

Zuru Tech India Pvt. Ltd., Ahmedabad, India

- Architected the core backend for an Influencer Management Platform handling content management, analytics, and media distribution using Node.js, PostgreSQL, and SQLite in a production multi-database setup.
- Developed an AWS Lambda API for automated media preview generation, cutting compute costs ~20% and improving media load times ~30%.
- Integrated YouTube Analytics API for real-time influencer performance tracking with automated data syncing across multiple channels.
- Optimized DB schema and query patterns, improving overall application performance by 20% and reducing query execution time by 40%.

PROJECTS

Netra: Influencer Management Platform (*Zuru Tech India*)

08/2023 – Present

- Engineered a centralized Express.js middleware layer with JWT/session auth, request validation, and dual-layer rate limiting: independently throttling inbound API traffic and outbound YouTube API quota to enforce hard quota boundaries without cross-interference.
- Implemented BullMQ + Redis distributed job scheduling with deduplication, retry logic, and concurrency control; built RabbitMQ consumer pipelines and cron-based schedulers for automated media publishing across Instagram, TikTok, and YouTube, fully decoupling heavy operations from the request lifecycle.
- Built a full asset library with watermark-based vendor distribution: background jobs produce processed copies with 6 placement options, proportional scaling, and edge-anchored positioning across image and video, keeping internal originals unchanged.

URL Shortener (*Infra-Heavy Backend with UI*)

github | live

- Built a URL shortening service with fire-and-forget click analytics via RabbitMQ: redirects publish to a queue without awaiting the write, keeping redirect latency independent of analytics throughput; failed messages route to a dead-letter queue for inspection.
- Implemented Redis in four distinct patterns: LRU cache (1h TTL) to skip Postgres on redirects, atomic INCR for concurrent-safe click counting, sorted-set sliding window for per-IP rate limiting, and a TTL-based jti blocklist for immediate JWT revocation, paired with a two-token auth system (15-min access + 7-day refresh) with per-token invalidation, role-based admin controls, and instant user banning enforced at the middleware level.
- Containerized the full stack (API, worker, Postgres, Redis, RabbitMQ) with Docker Compose for local development; deployed to Railway across 5 services on a private internal network with CI via GitHub Actions running an integration health check on every push.

AI/LLM Systems (*LLM Search Platform & Voice-Driven Coding Assistant*)

2025 – 2026 | github.com/dhrumilbhut

- Built a production-grade RAG pipeline with Python, PostgreSQL, and pgvector: Airflow-orchestrated ingestion and embedding generation with idempotent tasks, retries, and full pipeline observability in a containerized environment.
- Implemented metric-driven model governance with MLflow Model Registry, automatically promoting models to production based on retrieval quality, generation quality, and latency, preventing silent regressions.
- Architected a dual-server backend where a single core logic layer powers both a REST API and an MCP-compliant JSON-RPC server, eliminating code duplication and enabling flexible client integration.

SKILLS

Languages

Python | JavaScript | SQL

Backend & API

Node.js | Express.js | REST APIs | AWS Lambda | JWT | Session-based Auth | Middleware Design

Queues & Jobs

RabbitMQ | BullMQ | Redis | Dead-letter Queues | Cron Scheduling

Databases & Caching

PostgreSQL | SQLite | pgvector

Infrastructure & DevOps

Docker | Docker Compose | Git | GitHub Actions | AWS | Railway | Apache Airflow

Applied AI / LLM

RAG | Vector Search | LangChain | OpenAI API | Groq API | CrewAI | MLflow | Model Registry | Eval

Pipelines

EDUCATION

B.E. Computer Science & Engineering

2020 – 2024

Darshan Institute of Engineering and Technology, Rajkot, India

CGPA: 8.74

CERTIFICATES

IBM AI Engineering, IBM (Coursera)

Machine Learning Specialization, deeplearning.ai (Coursera)